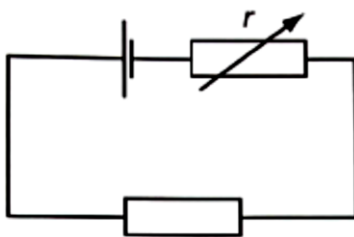
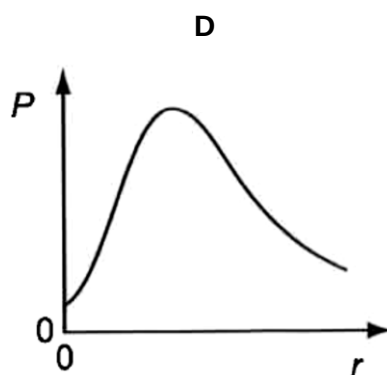
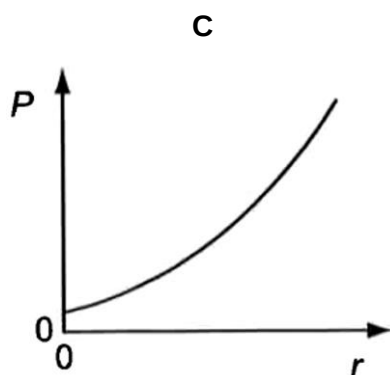
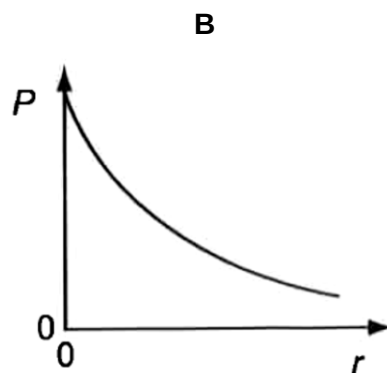
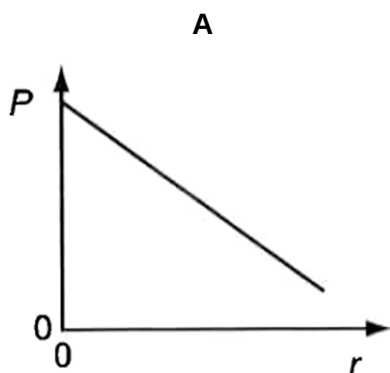


- 14 The circuit shown contains a cell of negligible internal resistance, a variable resistor of resistance r and a fixed resistor.



The resistance r is varied.

Which graph shows the variation of the power P dissipated in the fixed resistor as the resistance r of the variable resistor is changed?



- | | | |
|-----------|----------|--|
| 14 | B | $P = I^2 R = \frac{E^2 R}{(R+r)^2}$ <p>As r increase, I (current) in the circuit decreases, hence P decreases.
The equation shows this is not a linear relation</p> |
|-----------|----------|--|

15 A battery is connected to two lamps X and Y as shown